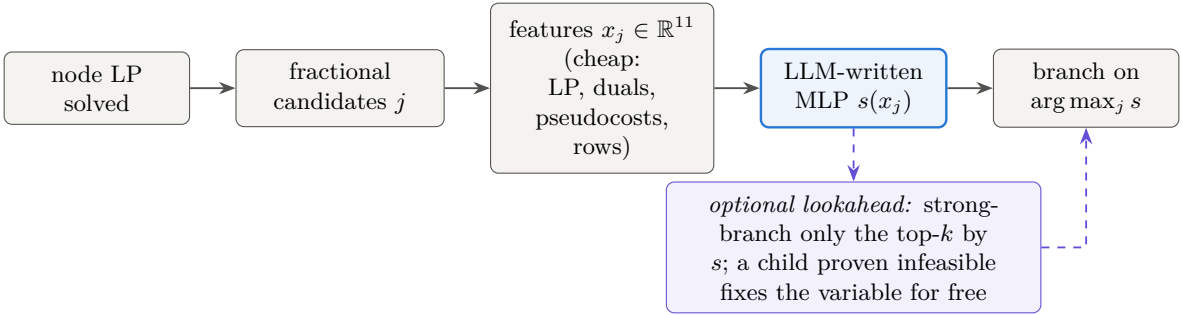

node LP solved
fractional candidates j
features $x_j \in \mathbb{R}^{11}$ (cheap: LP, duals, pseudocosts, rows)
LLM-written MLP $s(x_j)$
branch on $\arg\max_j s$
optional lookahead: strong-branch only the top-k by s; a child proven infeasible fixes the variable for free